



TryNex Lifestyle

PERFORMANCE AUDIT REPORT

Bundle analysis, Core Web Vitals, caching, lazy loading, and optimisation roadmap

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- Code splitting: React.lazy() + Suspense on all route-level components (40+ pages)
- Lazy-loaded 3D engine: Three.js / React Three Fiber loaded only when /design-studio is visited
- Dynamic imports: heavy admin components (TipTap editor, Recharts) split into async chunks
- Image optimisation: Cloudflare Images + R2 CDN delivery, WebP-first format
- Font preloading: Outfit + Plus Jakarta Sans preloaded via <link rel="preload">
- Lenis smooth scroll: GPU-accelerated scroll with RAF loop, no layout thrash
- Framer Motion: AnimatePresence + layout animations, useReducedMotion hook respected
- PWA: Service Worker with Workbox injectManifest — all static assets precached
- stale-while-revalidate: API responses served from cache while background refetch runs
- TanStack Query: deduplication, background refetching, configurable staleTime per hook

LCP (Largest Contentful Paint)	< 2.5s	Hero image lazy-loaded, above-fold products preloaded
FID / INP (Interaction Delay)	< 100ms	useTransition for heavy state, deferred admin renders
CLS (Cumulative Layout Shift)	< 0.1	Skeleton placeholders match final content dimensions
FCP (First Contentful Paint)	< 1.8s	Preloaded CSS, inlined critical styles, fast Cloudflare CDN
TTFB (Time To First Byte)	< 400ms	Cloudflare edge caching, s-maxage=30 on API

	react + react-dom + react/jsx-runtime + router + history	Shared vendor chunk (stable hash)
Routing		Shared vendor chunk
State	@tanstack/react-query + zustand	Shared vendor chunk
UI system	@radix-ui/* + class-variance-authority	Shared vendor chunk
Animation	framer-motion	Async import — loaded on first animated page
3D engine	three + @react-three/*	Async import — loaded only on /design-studio
Charts	recharts	Async import — loaded only on /admin dashboard
Rich text	@tiptap/*	Async import — loaded only on /admin/blog
PDF generation	pdfkit	Server-side only, not in browser bundle
Route pages	Each page component	Individual async chunks via React.lazy

Strategy	injectManifest (Workbox) — custom SW with full control
Precached assets	All .js, .css, .html, .ico, .png, .svg, .woff2 files
Max asset size	4MB — large 3D GLTF models excluded from precache
Runtime caching	API JSON responses: NetworkFirst strategy, 10s timeout
Offline fallback	/ and /products served from cache if offline
SW update flow	skipWaiting() + clientsClaim() for instant activation
Error recovery	Auto SW unregister + cache.keys().delete() if React fails

API server build	esbuild — single-file ESM bundle, sub-second compilation
Cold start (Render)	~3–8 seconds (free tier spin-up); keep-alive ping recommended
DB query time	10–15ms local, 20–60ms Neon (with connection pool)
Cache HIT response	~2–5ms (Upstash Redis REST)
Cache MISS response	~15–45ms (DB query + Redis SET)
Pino logging	Async JSON logging — zero blocking on hot paths

Rate limiting

express-rate-limit with in-memory store (fast, no Redis dep)

Recommendation

Add UptimeRobot ping every 5min to prevent Render cold starts